HMM-IDS for DDoS Detection: Ordered-Stream Report for the CICIDS2017 Friday Afternoon Trace

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Abstract: This work presents a temporal Hidden Markov Model (HMM)-based intrusion detection system evaluated on the CICIDS2017 DDoS dataset. After preprocessing, a balanced subset of 6,000 flow samples (50% Normal, 50% Attack) is constructed, with features transformed via $\log(1 + x)$ scaling and discretized into 64 symbols using K-Means. The HMM employs three hidden states—Normal, Suspicious, and Attack—where the Suspicious state captures intermediate behavior not explicitly labeled in the data. Unlike conventional shuffled evaluation, the model is trained and tested on an ordered data stream using a chronological split, enabling a more realistic temporal assessment. Parameters are initialized through supervised estimation and refined via Baum–Welch, with early stopping guided by validation Attack recall. Results on the ordered test stream show that the proposed approach achieves near-perfect Attack detection, improving Attack F1-score compared to a Random Forest baseline, while introducing a latent state that reduces forced misclassification of ambiguous Normal samples, highlighting the effectiveness of temporal modeling for network intrusion detection in realistic streaming scenarios.

Keywords: Hidden Markov Model, DDoS, Intrusion Detection System, CICIDS2017, Baum–Welch, Viterbi, K-Means, Random Forest, ordered-stream evaluation.

1. Introduction

A Distributed Denial-of-Service (DDoS) attack is a cyberattack in which multiple compromised machines simultaneously send a large volume of traffic to a target system, overwhelming its resources and preventing legitimate users from accessing the service. Such attacks typically evolve over time, exhibiting structured phases rather than isolated independent events. Distributed Denial-of-Service attacks therefore naturally present temporal dynamics, which makes them suitable for sequential modeling approaches.

This is exactly the kind of setting for which Hidden Markov Models (HMMs) are attractive: an HMM relates instantaneous observations to a latent traffic regime and encodes persistence through a transition matrix [1]. In the present project, the intended hidden-state interpretation is *Normal*, *Suspicious*, and *Attack*.

The central methodological update of this version is that the HMM is no longer trained and evaluated on a shuffled sample order. In the current code, the data loader first constructs an ordered sequence by interleaving 100-sample chunks from the BENIGN and DDoS pools, and the train/validation/test split is then performed chronologically with `shuffle=False`. This change matters because the transition matrix now operates on an ordered stream rather than on a random permutation, which makes sequence modeling substantially more defensible. At the same time, the evaluated sequence is still a *constructed ordered stream*, not the untouched raw chronology of the original network trace. The report therefore avoids calling the experiment a raw packet-level evaluation [3].

A second important clarification is that the deployed system is still *not* a supervised three-class classifier. The working dataset contains ground-truth labels only for Normal and Attack. The intermediate Suspicious state exists inside the HMM topology, but it receives no direct semantic supervision. The resulting model is therefore hybrid: states 0 and 2 are label-anchored, while state 1 remains a latent uncertainty compartment.

The goal of this report is to present the temporal pipeline, update all numerical results to the latest verified run, and describe the remaining limitations without overstating what the experiment proves. In particular, the report distinguishes the full Baum–Welch likelihood trajectory from the checkpoint that is actually selected by validation-guided early stopping.

2. Dataset and preprocessing

The experiment uses the CICIDS2017 file `Friday-WorkingHours-Afternoon-DDos.pcap\_ISCX.csv`. According to the official CICIDS2017 documentation, the CSV files are flow-level outputs generated by CICFlowMeter and labeled according to time, addresses, ports, protocols, and attack periods [2, 3]. The raw file contains 225 745 rows.

After cleaning invalid values, the pipeline constructs a balanced 6 000-sample working subset by taking 3 000 BENIGN flows and 3 000 DDoS flows. These flows are then arranged into an ordered sequence by alternating 100-sample chunks from each class pool. The final working subset therefore preserves local order within each extracted chunk, while also enforcing a balanced class structure across the ordered stream. No Suspicious labels are present in this curated subset.

Seven features are retained: *Flow Packets/s*, *Flow Duration*,

Table 1: Dataset and preprocessing summary for the temporal run

Element	Value
Dataset file	`Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv`
Raw rows	225 745
Effective dataset	6 000 flow samples
Class balance	3 000 Normal + 3 000 Attack
Retained features	7
Sequence construction	Ordered interleaving of 100-sample class chunks
Preprocessing	$\text{clip}(x, 0) \rightarrow \log(1 + x) \rightarrow$ RobustScaler
Discretization	K-Means, $k = 64$
Scaler fitting	Train only
K-Means fitting	Train only

Table 2: Train/validation/test split used in the temporal run

Set	Samples	Reported distribution
Train	4 200	Ordered slice (not stratified)
Validation	900	Ordered slice (not stratified)
Test	900	400 Normal + 500 Attack

Total Fwd Packets, *Bwd Packet Length Std*, *Flow IAT Max*, *Packet Length Variance*, and *Bwd IAT Mean*. The preprocessing pipeline applies $\text{clip}(x, 0) \rightarrow \log(1 + x)$ in the data loader, then fits a `RobustScaler` on the training slice only, and finally quantizes the scaled feature space into a 64-symbol alphabet by K-Means fitted on the training slice only. This ordering removes the leakage concern present in the earlier version of the project.

Table 1 summarizes the dataset and preprocessing pipeline used in the verified temporal run.

3. Experimental protocol

The temporal evaluation uses a chronological 70%/15%/15% hold-out split on the constructed ordered stream. The first 4 200 ordered samples are used for training, the next 900 for validation, and the final 900 for testing. Because the split is chronological rather than stratified, class proportions vary across slices. The final test stream contains 400 Normal samples and 500 Attack samples. Suspicious is absent from all labeled partitions.

This protocol differs from the previous shuffled version in two crucial ways. First, the HMM transition matrix is now learned and decoded on an ordered sample stream. Second, both the `RobustScaler` and K-Means codebook are fit strictly after the split, using only training data. The present results should therefore be interpreted as the outcome of an ordered, non-leaking evaluation on a curated 6 000-sample subset.

Three baselines are used for comparison. The first two are simple references: majority-class and random prediction. The third is a Random Forest trained on the scaled continuous feature vectors, without explicit sequence modeling. This provides a meaningful static-learning comparator for the HMM.

Table 3 summarize the ordered split and baseline results.

The validation slice is especially important because Baum–Welch is not selected by likelihood alone. The monitored criterion is validation Attack recall, which is aligned with the IDS objective of minimizing missed attacks.

Table 3: Baseline results on the final ordered test stream

Baseline	Accuracy	Attack F1
Majority-class	44.44%	0.00%
Random	52.00%	54.53%
Random Forest	97.89%	98.13%

4. Model and training procedure

The model is a discrete HMM with three hidden states and 64 observation symbols. The states are interpreted as state 0 = Normal, state 1 = Suspicious, and state 2 = Attack. The maximum number of Baum–Welch iterations is set to 20. In the current implementation, supervised initialization yields

$$\pi = [0.5,\ 0.0,\ 0.5].$$

Supervised initialization is also used to count labeled transitions and emissions on the training slice. In the verified run, the directly counted transition matrix before the manual "sweet spot" override is

$$A_{\text{supervised}} = \begin{bmatrix} 0.990 & 0.000 & 0.010 \\ 0.333 & 0.333 & 0.333 \\ 0.010 & 0.000 & 0.990 \end{bmatrix}.$$

The second row is uniform because no labeled transitions originate from the latent Suspicious state.

Before Baum–Welch starts, the Normal row is manually calibrated by setting $A_{0,0} = 0.95$, which changes the actual EM starting matrix to

$$A_{\text{start}} = \begin{bmatrix} 0.950 & 0.025 & 0.025 \\ 0.333 & 0.333 & 0.333 \\ 0.010 & 0.000 & 0.990 \end{bmatrix}.$$

The implementation also enforces a ceiling on Suspicious self-persistence, $A_{1,1} \leq 0.40$, in order to keep the unlabeled latent state from becoming a long-lived sink during EM refinement.

Baum–Welch is then run on the training observation sequence while monitoring validation Attack recall. The key point is that the EM trajectory and the deployed model are not the same object. On the monitored EM path, the training log-likelihood evolves as follows:

Checkpoint	Train log-likelihood	Validation Attack recall
Before BW	−12468.6881	—
Iter 1	−12468.8142	1.0000
Iter 2	−12345.0281	1.0000
Iter 3	−12273.9991	1.0000

Because validation Attack recall does not improve after its first optimum, early stopping restores the first recall-optimal checkpoint, i.e., the parameter set obtained at iteration 1. Therefore, the final deployed model is *not* the maximum-likelihood iterate from iteration 3. This distinction is essential for interpreting both the convergence figure and the final transition matrix.

Table 4: Final HMM-IDS metrics on the ordered test stream

Metric	Value
Global Accuracy	96.56%
Attack Recall	100.00%
Attack Precision	99.60%
Attack F1	99.80%
Normal Recall	92.25%
Normal Precision	100.00%
Predicted Suspicious rate	3.22%

Table 5: Exact confusion matrix (rows = real, columns = predicted)

	Pred-Normal	Pred-Suspicious	Pred-Attack
Real Normal	369	29	2
Real Suspicious	0	0	0
Real Attack	0	0	500

5. Main results

Table 4 reports the final HMM-IDS metrics on the 900-sample ordered test stream. The most important result from the IDS perspective is the Attack recall of 100.00%, which means that no Attack flow in the test stream is missed. Attack precision remains extremely high at 99.60%, yielding an Attack F1 of 99.80%.

However, the temporal HMM does not maximize every metric simultaneously. Its global accuracy is 96.56%, which is lower than the 97.89% achieved by the Random Forest baseline. The reason is visible in the confusion matrix: the HMM routes 29 Normal samples into the latent Suspicious state instead of forcing every observation into a binary Normal/Attack decision. In other words, the temporal model improves the attack-oriented metric of interest while paying an accuracy cost on Normal traffic due to explicit uncertainty handling.

The exact confusion matrix is given in Table 5. Out of 900 test samples, 869 are classified exactly on the diagonal. There are only two direct false alarms from Normal to Attack, zero missed attacks, and 29 outputs into the latent Suspicious state. Since the test set contains no Suspicious labels, those 29 samples count as off-diagonal errors in the strict three-state confusion matrix, even though they may also be interpreted operationally as deferred or uncertain alerts.

Under the strict metric definition used here, only predictions into the Attack column count as Attack false positives, which is why Attack precision remains 99.60%. Operationally, however, a deployment might also escalate Suspicious outputs. Under that broader interpretation, 31 of the 400 Normal test flows (29 routed to Suspicious and 2 routed directly to Attack) become non-normal alerts.

Relative to the Random Forest baseline, the temporal HMM improves Attack F1 from 98.13% to 99.80% and raises Attack recall to 100.00%, but it does not beat the Random Forest on global accuracy. The correct interpretation is therefore nuanced: the temporal HMM is better at the attack-detection objective emphasized here, while the Random Forest remains stronger on strict overall accuracy.

6. Interpretation of the diagnostic figures

6.1. Baum–Welch convergence

Figure 1 shows the full monitored Baum–Welch trajectory. The plot title reports a gain of about +194.7 nats (1.6%) because it compares the pre-BW log-likelihood to the final iterate reached by the EM loop at iteration 3. This gain belongs to the *trajectory*, not to the restored deployed model.

This figure should therefore be read in two layers. First, it confirms that the EM path itself is numerically well behaved. Second, it shows that the model-selection criterion is external to likelihood: the selected checkpoint is chosen for validation behavior, not because it maximizes the training objective. In the current implementation, the training log-likelihood is monitored along the loop, but the report does not treat the restored checkpoint as a separately recomputed maximum-likelihood solution.

6.2. Transition matrix before and after Baum–Welch

Figure 2 compares the actual EM starting matrix to the final selected transition matrix after early-stopping restoration.

The selected matrix is

$$A_{\text{selected}} = \begin{bmatrix} 0.977 & 0.016 & 0.008 \\ 0.467 & 0.400 & 0.133 \\ 0.010 & 0.000 & 0.990 \end{bmatrix}.$$

Several properties are worth highlighting. Normal and Attack are both highly persistent in the selected model, with self-transition probabilities 0.977 and 0.990, respectively. The Suspicious row hits the reported ceiling at $A_{1,1} = 0.400$, which confirms that the topological restriction is active rather than decorative. Its strongest outgoing transition is toward Normal (0.467), followed by self-persistence (0.400), then transition toward Attack (0.133). This makes the latent state behave more like a bounded buffer near the Normal regime than like a broad absorbing class.

6.3. Emission matrix analysis

Figure 3 presents the selected emission matrix after Baum–Welch restoration. Each subplot uses its own vertical scale, so the figure should be interpreted within each row rather than by direct comparison of bar heights across rows.

The Normal row is multi-modal and comparatively broad. In the displayed figure, the most visible labeled peaks occur around symbols #2, #13, #15, #22, and #24, which suggests that benign traffic still occupies several distinct regions of the discretized feature space. The Attack row is much sharper, with a dominant peak around symbol #17 and other emphasized peaks around #4, #5, #16, and #23. This concentration is consistent with the model's very high Attack precision and perfect Attack recall on the final test stream.

The Suspicious row remains much flatter than the Attack row. It is not perfectly uniform, but it is clearly less semantically anchored than the two supervised edge states. The combination of a flatter emission profile and a capped self-transition helps explain why Suspicious appears only in a minority of decoded test positions.

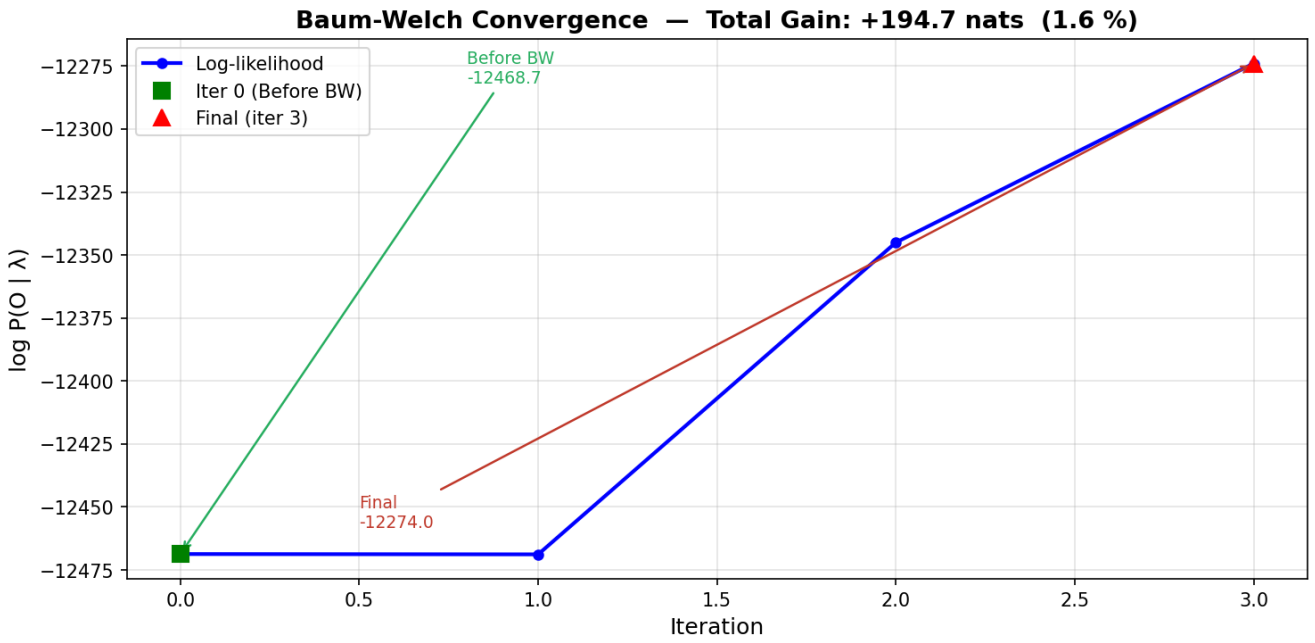


Figure 1: Monitored Baum–Welch trajectory on the training sequence. The plotted EM path rises from −12468.69 before BW to −12274.00 by iteration 3, for a trajectory gain of about +194.7 nats (1.6%). However, validation Attack recall reaches its optimum at the first checkpoint and does not improve afterwards, so early stopping restores the first recall-optimal model rather than the maximum-likelihood iterate.

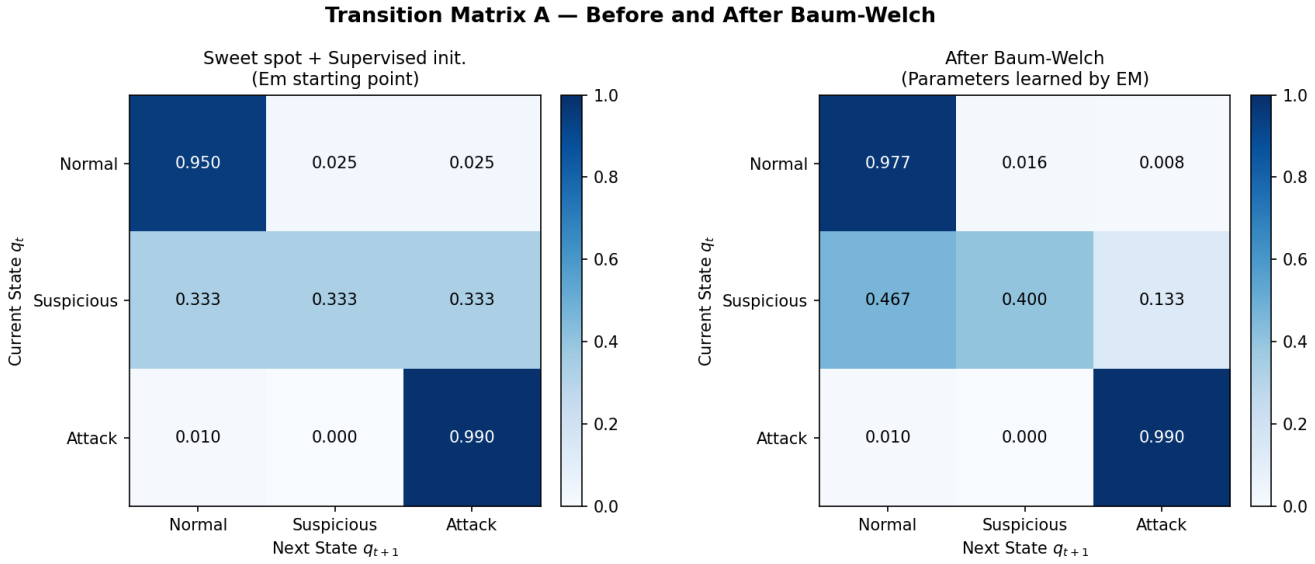


Figure 2: Transition matrix A before and after Baum–Welch. Left: the actual starting point of EM after the $A_{0,0} = 0.95$ sweet-spot override. Right: the selected transition matrix retained after validation-guided early stopping and restoration of the first recall-optimal checkpoint.

6.4. Confusion matrix heatmap

Figure 4 shows the confusion matrix in absolute counts and row-normalized percentages. Because Suspicious is absent from the real labels, the only directly meaningful recalls are those for Normal and Attack.

The most important pattern is that the Attack row is perfectly preserved: all 500 Attack flows are decoded as Attack. The dominant off-diagonal mass comes instead from 29 Normal flows routed into Suspicious. This is why the model achieves perfect sensitivity to attacks while not maximizing global accuracy.

6.5. Ordered-stream visualization

Figure 5 compares the real and decoded state sequences over the first 500 positions of the final ordered test stream. In this window, the figure reports 16 off-diagonal positions (3.2%) and 14 Suspicious predictions.

Unlike the older shuffled protocol, this plot is now meaningful as an ordered-sequence diagnostic. It shows that the decoded path closely tracks the large blocks present in the evaluation stream, while the few disagreements are sparse and concentrated near local transition regions. The presence of short orange segments in the prediction band confirms that the latent state is used as a local uncertainty mechanism rather than as a dominant alternative class.

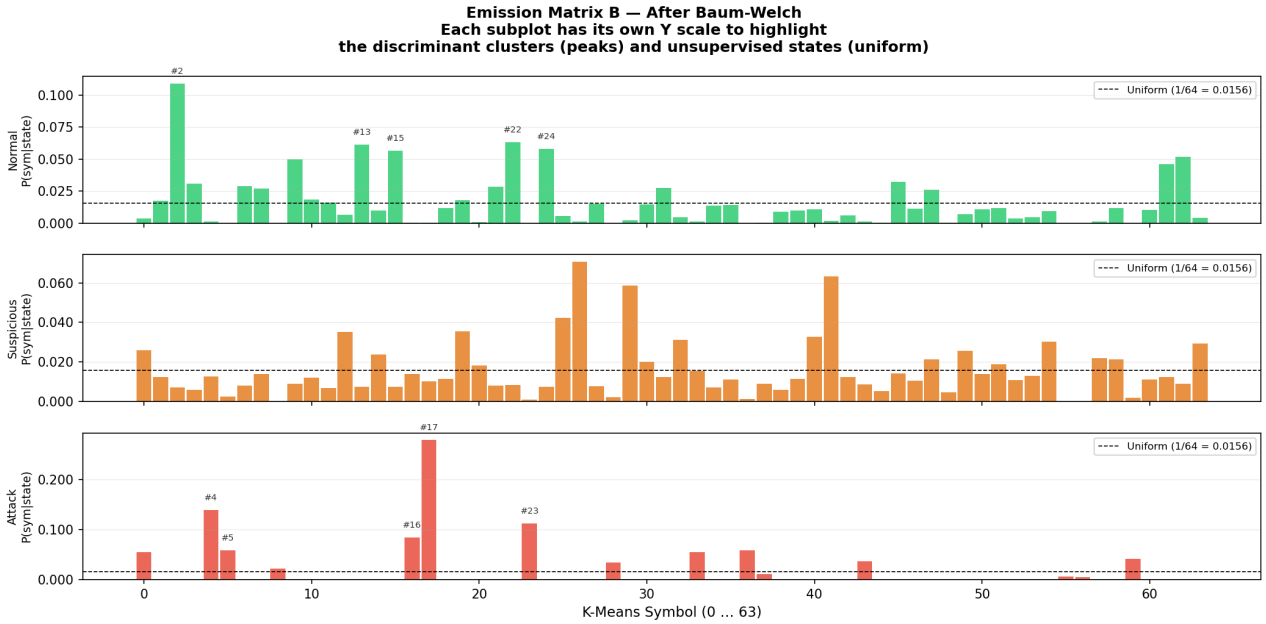


Figure 3: Selected emission matrix B after Baum–Welch restoration. The dashed line marks the uniform probability $1/64 = 0.0156$. Each subplot uses its own vertical scale to highlight discriminant peaks and the flatter structure of the latent state.

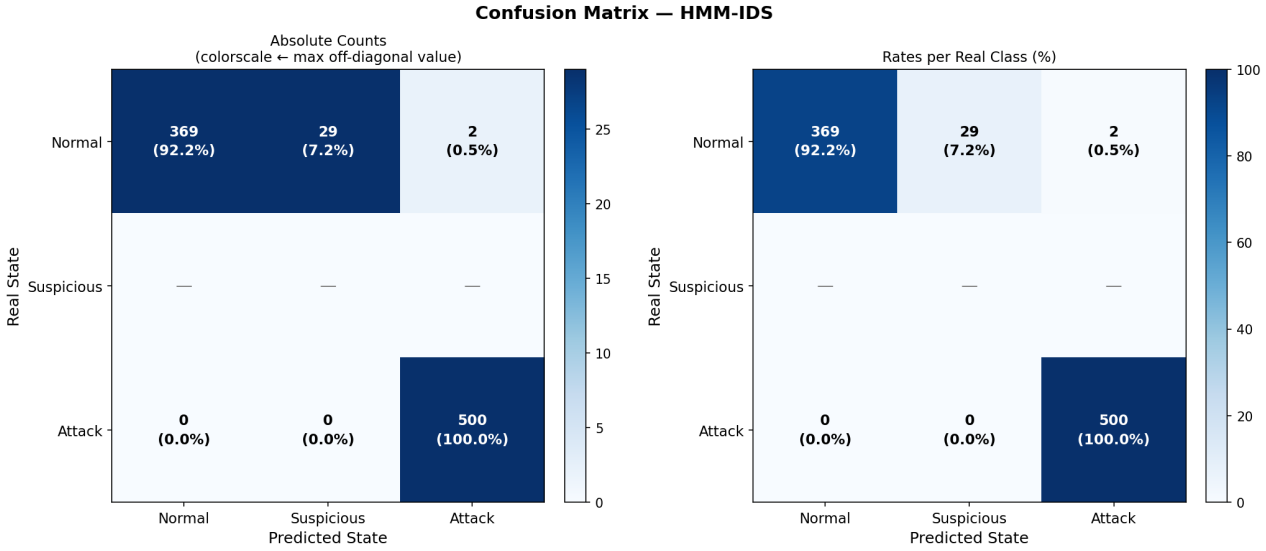


Figure 4: Confusion matrix of HMM-IDS on the ordered test stream. The left panel shows absolute counts and the right panel shows row-normalized percentages. Suspicious does not appear in the ground truth, so the informative recalls are 92.25% for Normal and 100.00% for Attack.

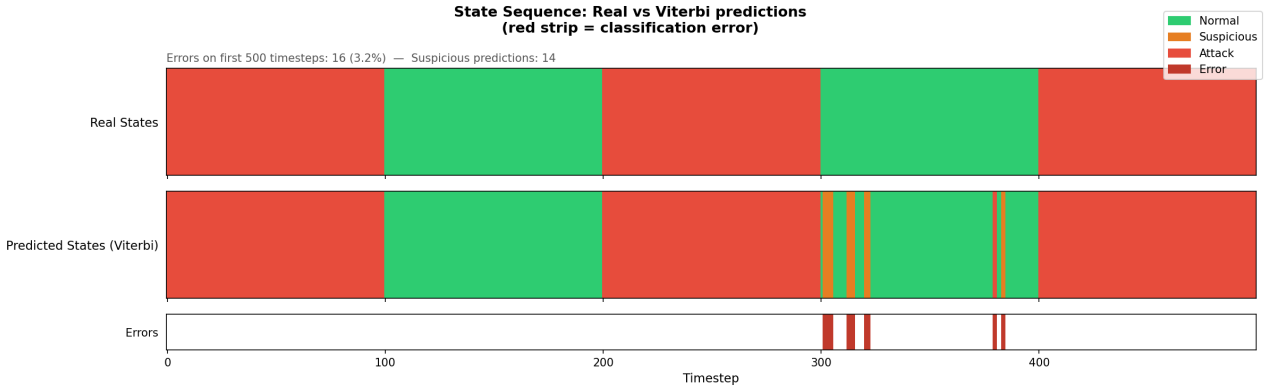


Figure 5: Ordered-stream comparison for the first 500 positions of the final test sequence. The top band shows real states, the middle band shows Viterbi predictions, and the bottom band marks off-diagonal positions. In this 500-sample window, the figure reports 16 off-diagonal outputs and 14 Suspicious predictions.

7. Discussion

The temporal results resolve several major issues from the earlier version of the project. The sequence is no longer shuffled, preprocessing is no longer fitted on the full dataset before splitting, and the report now distinguishes between the EM trajectory and the checkpoint that is actually selected for deployment. These are real methodological improvements.

At the same time, the results need to be interpreted carefully. First, the validation criterion is Attack recall, and it saturates at 1.0000 from the first Baum–Welch checkpoint onward. Because the tie is effectively broken by first attainment, the selected model is restored from iteration 1 even though later iterations achieve higher training likelihood. This means that the positive role of Baum–Welch in this run is mainly diagnostic and regularizing rather than a direct source of additional selected likelihood gain.

Second, the temporal HMM does not dominate the Random Forest on every metric. Its advantage is concentrated on the attack-detection objective: it eliminates attack misses and slightly improves Attack F1. However, it also routes some Normal flows into the latent Suspicious state, which lowers overall accuracy relative to the Random Forest. This trade-off may be acceptable in an IDS context if missed attacks are considered more costly than uncertain benign alerts, but it should be stated explicitly rather than hidden behind a single headline number.

Third, the three-state architecture works well here, but this report still does not prove that three hidden states are universally better than two. The evidence supports a narrower claim: on this curated ordered subset, a three-state HMM with a bounded latent middle state yields excellent attack-oriented performance and interpretable ordered-stream behavior. A direct ablation against a two-state HMM would still be needed to prove the added value of the intermediate state itself.

Finally, the experiment remains limited to a balanced 6 000-sample subset extracted from a much larger CSV file. The results are therefore valid for this specific ordered hold-out protocol and this curated subset, but they should not yet be generalized to full-dataset deployment without additional experiments.

8. Conclusion

This report aligns the HMM-IDS pipeline with the latest verified code, outputs, and figures. The main methodological corrections are now in place: ordered hold-out splitting with `shuffle=False`, train-only scaling, train-only K-Means fitting, explicit separation between the EM trajectory and the restored checkpoint, and a more careful interpretation of the latent Suspicious state.

On the final 900-sample ordered test stream, the HMM reaches 96.56% global accuracy, 100.00% Attack recall, 99.60% Attack precision, and 99.80% Attack F1. The Random Forest baseline remains stronger on global accuracy (97.89%), but the HMM improves the attack-oriented objective by eliminating false negatives and raising Attack F1 from 98.13% to 99.80%.

The main technical lesson is therefore specific and defensible. In this ordered, interleaved subset of CICIDS2017, a discrete three-state HMM with a bounded latent intermediate state is consistent with a benefit from ordered context for attack-sensitive detection, provided that sequence ordering is preserved and preprocessing leakage is avoided. The model is best understood not as a universal winner on every metric, but as a temporal IDS prototype that trades some Normal-class exact-match accuracy for stronger attack detection and an explicit uncertainty mechanism.

References

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